Summations exercises

All questions are original. Index and empty-set conventions follow the lecture.

A-E01 Arithmetic and square sums with shifted starts

Compute $\sum_{k=4}^{9}(3k-2)$ and $\sum_{k=2}^{5}k^2$ in two ways. Derive the square-sum formula by telescoping $(k+1)^3-k^3$ rather than quoting it.

A-E02 Finite geometric sums and error

Find $\sum_{k=0}^{5}(-1/3)^k$, its infinite sum, and the truncation error. Explain separately why $x=1$ and $x=-1$ cannot use the infinite formula. Derive a finite formula for $\sum_{k=0}^{n}kx^k$ before taking a valid limit.

A-E03 Telescoping products and logarithm premises

Find $\sum_{k=1}^{n}1/[(k+1)(k+3)]$ and $\prod_{k=2}^{n}(1-1/k^2)$. Check both at $n=3$. When is conversion to real logarithms illegal?

A-E04 Induction with fixed constants

For $S_n=\sum_{k=0}^{n}5^k$, find a fixed constant c proving $S_n\le c5^n$, including the base case $n=0$. Give a lower bound that proves a Θ result. Identify exactly how the constant fails in the claim that $\sum_{k=1}^{n}k=O(n)$ remains linear after adding $n+1$.

A-E05 Ratios and prefixes

For $\sum_{k=1}^{\infty}k^2/4^k$, find a fixed ratio bound $\rho<1$ valid from $k=2$, and obtain a finite upper bound. Then compute the exact sum using weighted geometric series. Why are ratios below 1 insufficient in general?

A-E06 Two harmonic brackets

At $n=12$, list the dyadic blocks and each block's exact sum or fraction expression. Give logarithmic upper and lower bounds. Derive $\ln(n+1)\le H_n\le 1+\ln n$ from unit intervals. Why can the upper bound not use the integral of $1/x$ over $[0,n]$?

A-E07 p-series and changing parameters

Use integration to prove convergence for a fixed $p>1$, and give an explicit bound when $p=3$. Give the Θ order for $0<p<1$. Why is $1+1/(p-1)$ not independent of n when $p=1+1/n$?

Summations exercises (continued)

A-E08 Nested loops and index sets

Compute $\sum_{i=1}^{n} \sum_{j=i}^{n} (2j + 1)$ by finite reordering to obtain an exact polynomial. Check it at $n = 3$. If each cost is $\Theta(j)$, what uniformity condition is implicit in applying Θ to the whole sum?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study

Lin Cing
https://github.com/xinhi6716/computer-science-self-study